Adiel Krisnanda Franstito

Supply Chain Delivery Performance & Financial Risk Analysis

Identifying Systemic Delivery Delays and Their Financial Impact

Executive Summary

Bullet Points:

- **57% of total orders experienced late delivery**, indicating a **systemic supply chain issue** rather than a regional problem.
- Late delivery rates remain consistent across **markets and regions, suggesting structural inefficiencies**.
- Moderate and high-risk delays generate **disproportionate revenue loss**, despite lower order volume.
- Faster shipping modes **reduce delay duration** but **fail to significantly reduce late delivery rates**.
- A severity-based prioritization strategy presents the highest opportunity to **reduce financial risk**.

Business Problem & Objectives

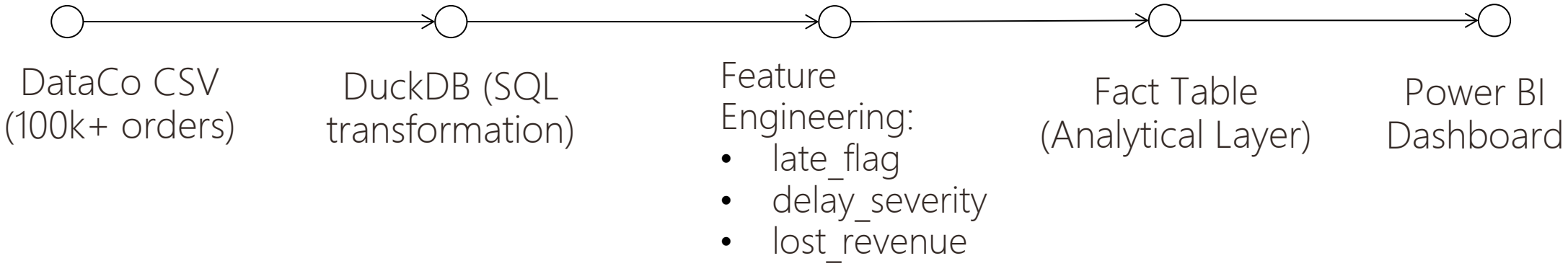
- Business Problem

High late-delivery rates negatively impacting customer experience and revenue.
Limited visibility into which delays actually drive financial risk.
Operational decisions are often based on volume, not severity or impact.

- Project Objectives

Identify key drivers of delivery delays.
Quantify financial impact of delay severity.
Provide actionable, data-driven recommendations.

Data Architecture & Analytical Framework



Analytical Methodology & KPI Design

01. Data Preparation

- SQL-based data cleaning using DuckDB
- Date standardization via STRPTIME
- Removal of null shipping records

02. Feature Engineering

- Late Delivery %
- Revenue-at-Risk %
- Severity-based risk segmentation
- Market & shipping mode comparison

03. KPI Modeling

- Delivery delay calculation
- Binary late_flag indicator
- Delay severity classification (Mild, Moderate, Severe)
- Revenue-at-risk calculation (conditional aggregation)

04. Analytical Framework

- Volume Analysis
- Severity Analysis
- Financial Impact Analysis

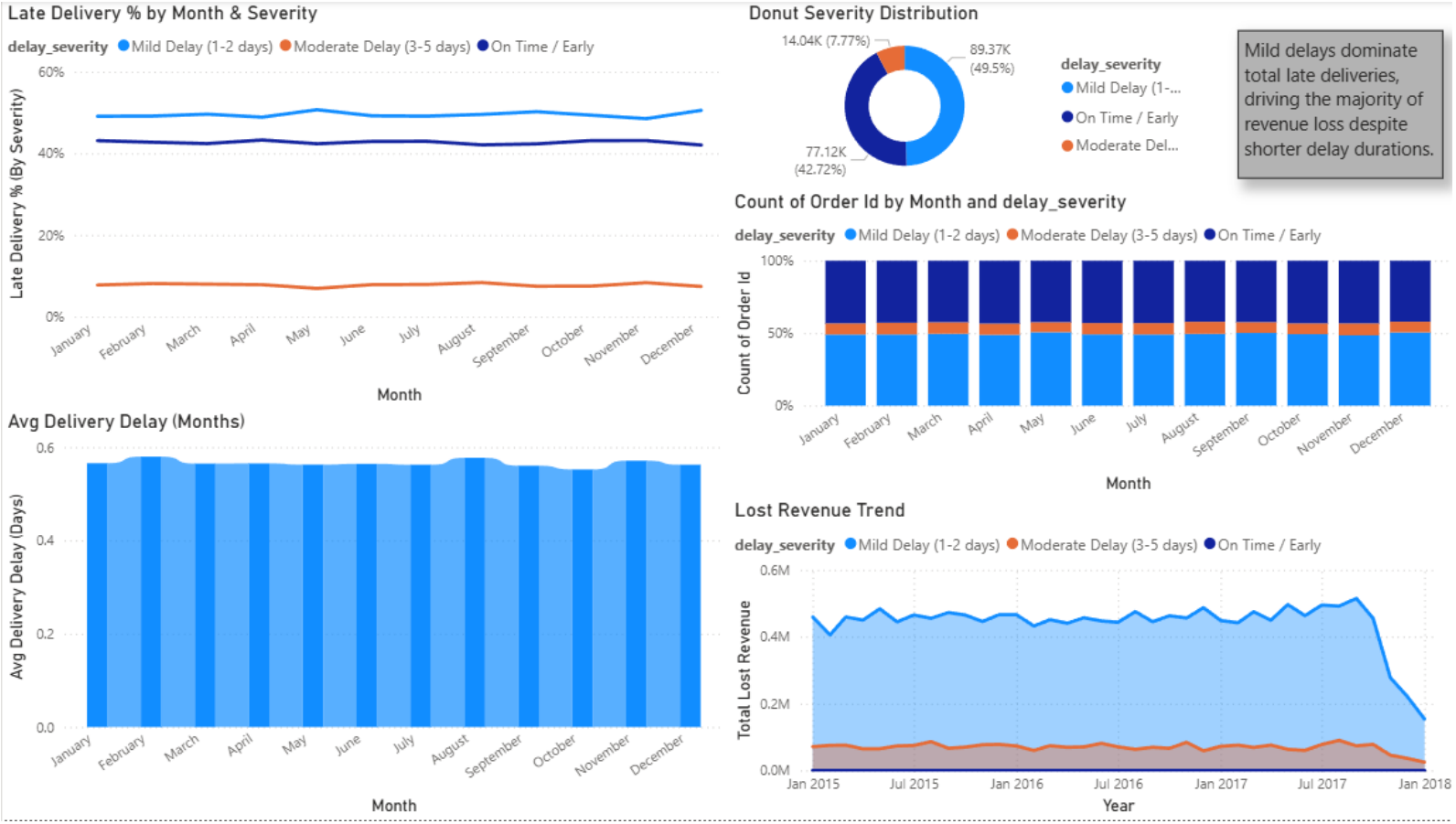
Dataset Overview

Dataset : Order-level delivery

Key attributes:
- Market & region
- Shipping mode
- Delivery delay (days)
- Delay severity classification
- Sales & lost revenue

Analytical Approach :
- KPI design using DAX (Late Delivery %, Lost Revenue, Risk Rate)
- Segmentation by **market, region, shipping mode**
- Risk classification based on delay severity
- Interactive analysis using **Power BI**

Delivery Performance Overview

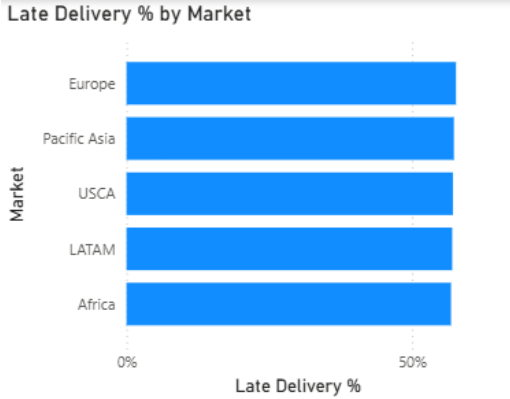


Delivery delays remain consistently high across time, with mild delays dominating in volume but not necessarily in risk.

Key Takeaways:

- Late deliveries account for the majority of total orders.
- Volume alone does not reflect financial or operational risk.
- Deeper severity analysis is required.

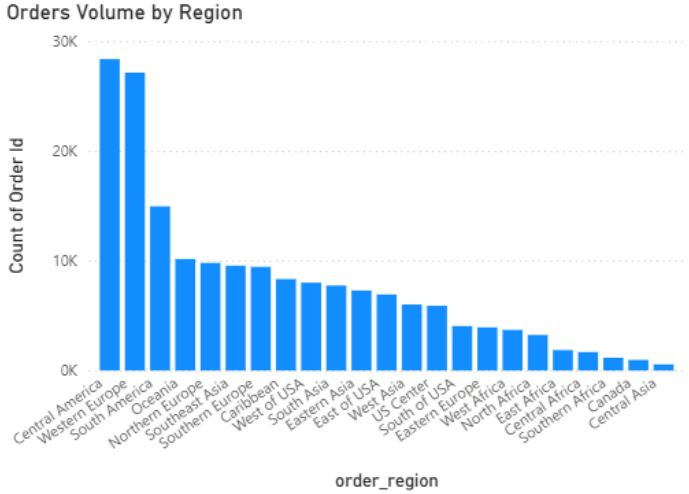
Market & Shipping Mode Insights



Market	Total Orders	Late Orders	Late Delivery %
Africa	11614	6598	56.81%
Europe	50252	28989	57.69%
LATAM	51594	29420	57.02%
Pacific Asia	41260	23649	57.32%
USCA	25799	14744	57.15%
Total	**180519**	**103400**	**57.28%**

57.28%

Late Delivery %



"Although the late delivery percentage is similar across all markets, the business impact is not the same. LATAM and Europe generate the highest number of late orders due to their large order volumes. Therefore, improvement efforts should prioritize high-volume markets rather than regions with slightly higher percentages."

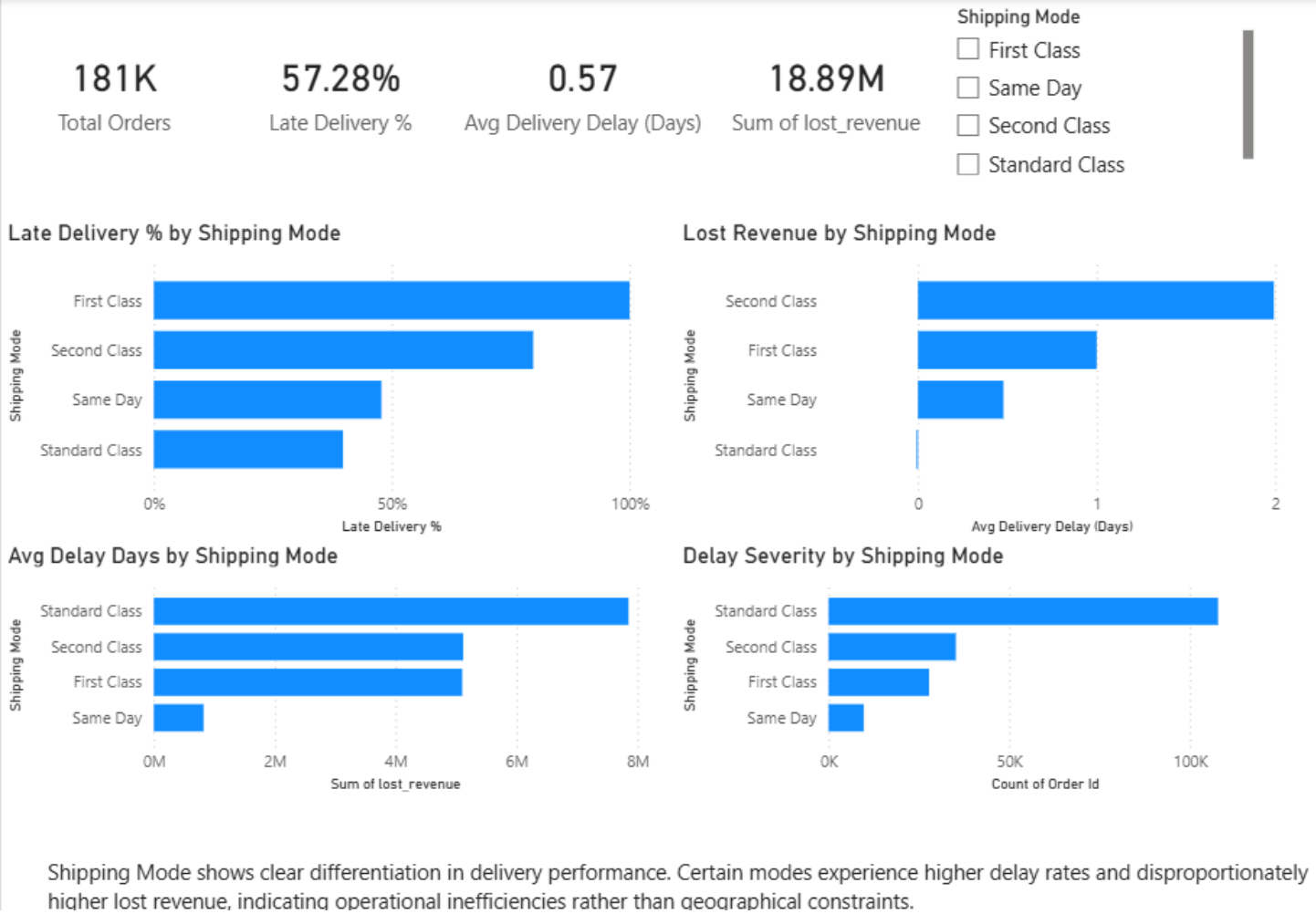
The late delivery rate remains consistently high at around 57% in all regions. This indicates that delivery delays are driven by systemic issues rather than market-specific factors.

Key Insights:

- Late delivery is not market-specific.
- Regional differences are minimal.
- Premium shipping modes reduce delay days but not late-delivery probability.

Market and shipping speed alone do not explain delivery performance issues.

Market & Shipping Mode Insights

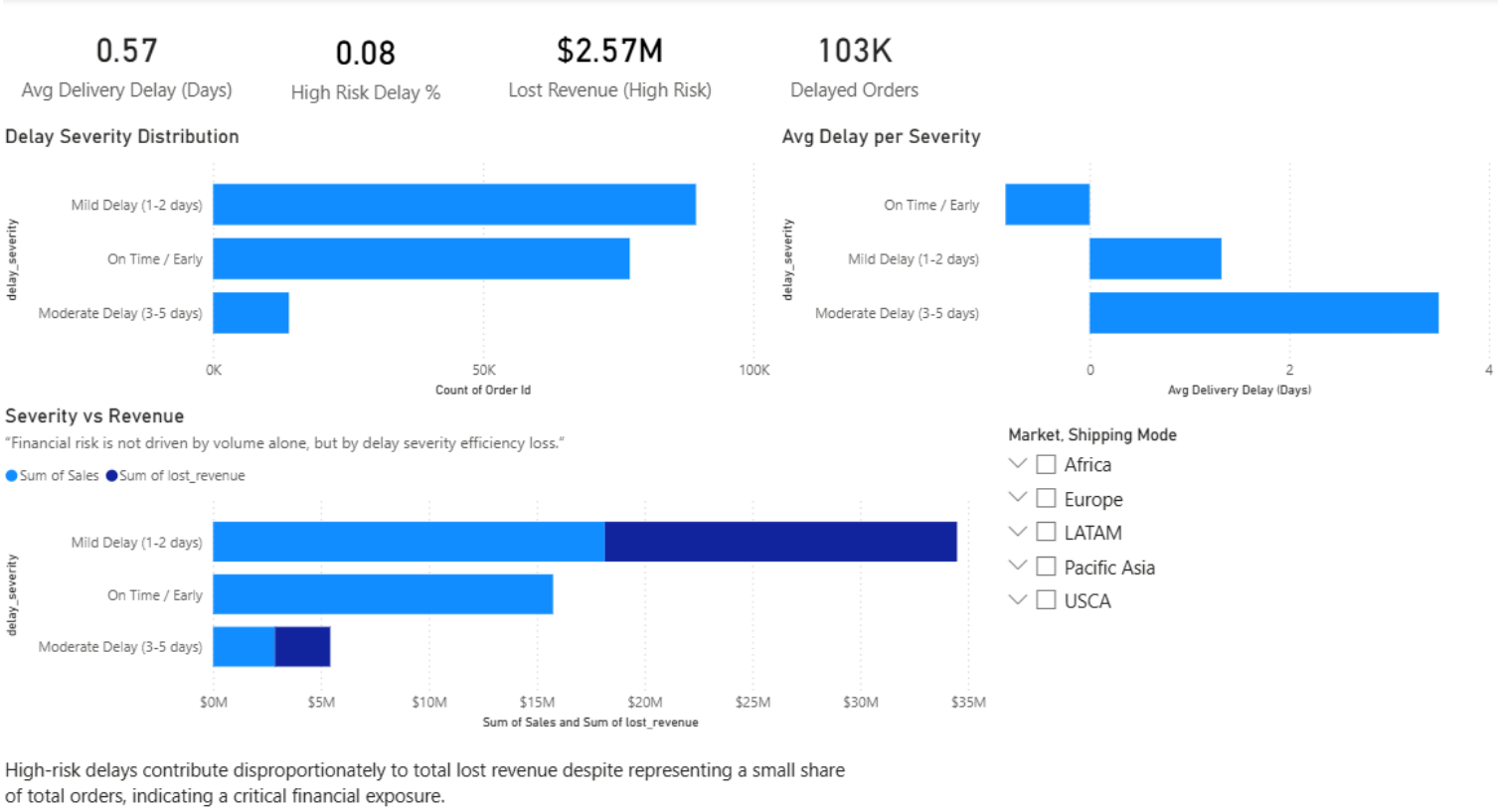


Market and shipping speed alone do not explain delivery performance issues.

Key Insights:

- Standard class and Second class make the highest lost revenue
- Same day made the lowest lost revenue

Delivery Risk & Financial Impact



High-risk delays contribute disproportionately to total lost revenue despite representing a small share of total orders, indicating a critical financial exposure.

Key Insights:

- Moderate and high-risk delays contribute significantly to lost revenue.
- Financial impact is driven by severity, not frequency.
- Small portions of orders create outsized risk.

Key Business Insights

01. Delivery delays are systemic across the supply chain

02. Revenue loss is concentrated in higher-severity delays

03. Improving speed without workflow optimization offers limited benefit

04. Risk-based analysis provides clearer decision support

Recommendations & Action Plan

Recommended Actions:

- Prioritize mitigation of moderate and high-risk delays.
- Re-evaluate shipping workflows, including premium services.
- Implement early-warning indicators based on delay severity.
- Align operational KPIs with financial risk exposure

Expected Business Impact:

- Reduced lost revenue
- Improved delivery reliability
- Better operational prioritization

Technical & Analytical Challenges

Challenges	Solutions
Large dataset handling	Used DuckDB for in-memory SQL processing
Date parsing inconsistencies	Standardized date transformation via STRPTIME
Defining revenue-at-risk metric	Developed severity-based conditional revenue model

Tools, Skills & Value Delivered

Technical Skills Demonstrated

- SQL-based data transformation
- Feature engineering
- Fact table design
- KPI & DAX modeling
- Financial risk analysis

Value Delivered

- Identified systemic delay drivers
- Quantified revenue exposure
- Developed severity-based prioritization model

Thank You